

## The Benefit Cost Ratio Method

Benefit Cost Ratio B/C or BCR or also called Benefit Cost Analysis (BCA). It is a technique that is widely used to assess or evaluate an alternate or a project to find out whether benefits are higher than costs or vice - versa. B/C or BCR or BCA is method to find out PV, annual cost and ROR. B/C provides a means for comparing benefits to costs.

The objective of Chapter on Benefit Cost Ratio is how to compare 2 alternatives on the basis of benefit/cost ratio. This method is required as a technique.

The higher the benefits the better

$$B/C = AWB/AWC$$

$B/C > 1$  Project desirable.

Methods of determining B/C

1. Conventional Method
2. Modified Method

### Conventional Method

Conventional Method is older than Modified Method

In conventional method there is it is not possible to calculate Dis-benefit (DB)

As a first step calculate value of CR and then find value of B/C ratio

$$B/C = B / CR + O\&M \text{ ----- (1)}$$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{ ----- (2)}$$

Where

B = Annual Worth Benefit {AWB}

CR = Capital Recovery Cost – Initial investment converted to annual cost.

O&M = Operation & Maintenance Expenses of proposed project  
 F = Future Worth or Salvage Value (SV)  $B/C > 1$  Project  
 Desirable.

## Modified Method

Modified Method was derived later

In Modified Method there is provision to calculate DB Equation:

$$B/C = B - DB - O\&M / CR \text{ -----}(3)$$

If DB is zero then equation would be

$$B/C = B - (O\&M) \text{-----}(3a)$$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{ -----} (4)$$

Where

B = Annual Worth Benefit (AWB)

CR = Capital Recovery Cost

O&M = Operation & Maintenance Costs

DB = Annual Dis-benefit

If there is no Disbenefit (DB) then equation will change:

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{ -----}(4)$$

= Future Worth

F

$B/C > 1$  Project is Desirable

### Problem 1

Determine the B/C ratio by Conventional Method for a project with the following details:

|                 | <u>Alternate A</u> | <u>Alternate B</u> | <u>Initial</u> |
|-----------------|--------------------|--------------------|----------------|
| cost            | \$ 20,000          | \$ 24,000          |                |
| Project life    | 5 years            | 5 years            |                |
| SV              | \$ 4,000           | \$ 4,000           |                |
| Annual benefits | \$ 10,000          | \$ 10,000          |                |
| Annual O&M      | \$ 4,400           | \$ 4,400           | Interest       |
| rate            | 8%                 | 8%                 |                |

### Solution:

Given:

|                        |           |
|------------------------|-----------|
| Initial Cost I         | \$ 20,000 |
| Project Life N         | 5 years   |
| Future Worth For SV    | \$ 4,000  |
| Annual Worth Benefit B | \$ 10,000 |
| Annual O&M             | \$ 4,400  |
| Interest rate (i)      | 8%        |
| By Conventional Method |           |

Formula:

$$B/C = B / CR + O\&M \text{-----} \quad \text{Eq(1)}$$

$$CR = (I - F) [i(1+i)^N / (1 + i)^N - 1] + F(i) \text{--- Eq (2)}$$

Substituting values in Eq (2):

$$(20000 - 4000)[8\%(1 + 8\%)^5 / (1 + 0.8\%)^5 - 1] + 4000(8\%) \}$$

$$\begin{aligned}
CR &= (20,000 - 4000)\{0.08(1+0.08)^5 / (1+0.08)^5 - 1\} + 4000(0.08) \\
CR &= 16000 [0.08(1.08)^5 / (1.08)^5 - 1] + 4000(0.08) \\
CR &= 16000 [0.08(1.469)/1.469 - 1] + 320 \\
CR &= (0.1175/0.469) + 320 \\
CR &= 16000 (0.25) + 320 \\
CR &= 4000 + 320 \\
CR &= 4320
\end{aligned}$$

Substituting values in Eq (1)

$$B/C \text{ ratio} = 10,000 / 4,327 + 4400$$

$$10,000/8,727$$

$$B/C \text{ ratio} = 1.146 > 1.0$$

Problem 1 [Contd.]

By Modified Method DB being zero

$$\text{Formula } B/C = B - (O\&M) / CR \text{----- Eq (3)}$$

Substituting values in Eq(3)

Substituting value of CR from Conventional Method above

$$\begin{aligned}
B/C \text{ ratio} &= 10,000 - 4,400 / 4,327 \\
&= 1.294 > 1.0
\end{aligned}$$

B/C > 1.0 by either method the project is desirable and recommended  
**Problem 2**

- Describe Benefit and Disbenefits of construction a Dam projects.
- Determine the B/C ratio based on modified method of projects for following given values:

| <u>Details:</u>    | <u>Alternate A</u> | <u>Alternate B</u> |
|--------------------|--------------------|--------------------|
| Initial investment | \$ 300,000         | \$ 500,000         |
| Annual Income      | \$ 25,000          | \$ 25,000          |
| Annual O&M         | \$ 5,000           | \$ 5,000           |
| Annual Dis-benefit | \$ 2,000           | \$ 2,000           |

Project life  
SV or F  
Interest rate

25 years  
\$ 0  
10%

25 years  
\$ 0  
10%

### Solution:

Given:

Initial investment

\$ 500,000

10% N years

25

years

Annual Income (B)

\$ 25,000

DB

\$

2,000 Salvage

Value F

\$ 0

By Modified Method

Formula:  $B/C = B - DB - (O\&M) / CR$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i)$$

Substituting values in above equations

B/C

Project B Contd.

$$CR = (I - F) \left[ \frac{i\%(1 + 10\%)^n}{(1 + i\%)^n - 1} \right] + SV (i\%)$$

$$CR = (500,000 - 0) \left[ \frac{10\%(1 + 10\%)^{25}}{(1 + 10\%)^{25} - 1} \right] + 0 (10\%)$$

$$CR = 500,000 \left[ \frac{0.1}{(1.1)^{25} - 1} \right] \frac{(1.1)^{25}}{1}$$

$$CR = 500,000 \left[ \frac{0.1 (10.8)}{(10.8) - 1} \right]$$

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$$CR = 500,000 \left[ \frac{1.08}{9.8} \right]$$

$$CR = 500,000 (0.11)$$

$$CR = 55,000$$

$$B/C = \frac{B}{CR + O\&M}$$

$$B/C = \frac{25,000}{55,000 + 5000}$$

$$B/C = \frac{25,000}{55,000 - 5,000}$$

$$B/C = \frac{25,000}{60,000}$$

$$B/C = 0.41$$

Project being  $> 1$  is not desirable

$$B/C = (25000 - 2000 - 5000) / (500,000 - 0) [0.1(1.1)^{25} / (1.1)^{25} - 1] + 0$$

$$B/C = 18,000 / 500,000[0.11]$$

$$B/C = 18,000 / 55,000$$

$$B/C = 0.327$$

$0.327 < 1.0$  Project not desirable